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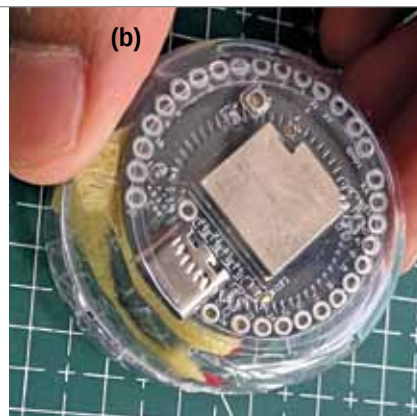
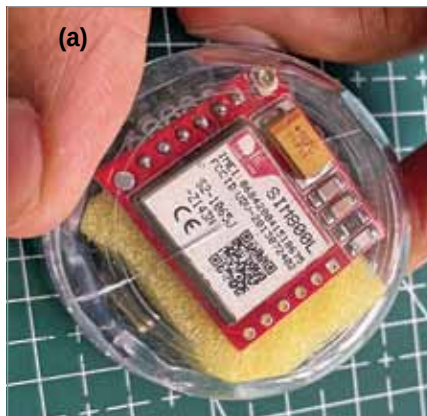


Fig. 6: Setting up the device in an enclosure: (a) SIM800L side, (b) IndusBoard side

location. The phone number can be set within the code.

Fig. 3 shows the code snippet for setting up the phone number.

Circuit design

The IndusBoard Coin acts as the central controller, interfacing with the SIM800L module for GSM communication. The microphone is connected to the SIM800L's audio input, allowing captured sound to be transmitted over a call.

Fig. 4 shows the circuit diagram for the long-range spy device.

Power connections must be carefully managed. The IndusBoard Coin operates at 3.3V, whereas the SIM800L module requires a stable 4V power supply. To achieve this, a voltage regulator can step down the Li-Po battery's voltage. The voltage regulator connects the battery to the IndusBoard Coin and the SIM800L module. The microphone module is connected to the SIM800L's MIC +

and MIC- pins to ensure precise audio transmission.

Note. If using the IndusBoard Coin V2.1, it has an in-built battery connector. The battery can be directly plugged in using the built-in BMS module on that connector.

Fig. 5 shows the battery connector on the IndusBoard Coin V2.1.

Construction and testing

After soldering the SIM800L, battery, and microphone, all components are fixed within a 3cm enclosure. The coin comes in a 3cm circular case, where the components are placed before sealing the entire setup. Five to seven small holes should be made in the case to allow sound to enter.

Fig. 6 shows how the device is set up within the enclosure.

A SIM card with a voice call pack must be inserted before charging the battery. The device requires a few seconds to connect to the network. When surveillance is needed, calling the configured phone number enables audio monitoring.

EFY note. This device is intended strictly for ethical security applications. Any unauthorised use for surveillance is illegal. **EFY**

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